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w/AMENDMENT 2
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DETAIL SPECIFICATION

# MATERIAL, BALL VALVE SEAT SEAL AND THRUST WASHER

This specification is approved for use by all Departments and Agencies of the Department of Defense.

## 1. SCOPE

1.1 Scope. This specification covers material used for ball valve seat seals and thrust washers (nylon 66 with molybdenum disulfide) in petroleum- and polyalphaolefin (PAO)-based hydraulic fluid systems.

## 2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 or 4 of this specification. This section does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 or 4 of this specification, whether or not they are listed.

2.2 Government documents.

2.2.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

FEDERAL SPECIFICATIONS

L-P-410 - Plastic, Polyamide (Nylon), Rigid: Rods, Tubes, Flats, Molded and Cast Parts

(Copies of this document are available online at http://quicksearch.dla.mil/.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME B46.1 - Surface Texture (Surface Roughness, Waviness and Lay)

(Copies of this document are available online at www.asme.org/.)

Comments, suggestions, or questions on this document should be addressed to Commander, Naval Sea Systems Command, ATTN: SEA 05S, 1333 Isaac Hull Avenue, SE, Stop 5160, Washington Navy Yard DC 20376-5160 or emailed to CommandStandards@navy.mil, with the subject line "Document Comment". Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.



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ASTM INTERNATIONAL

ASTM D785 - Standard Test Method for Rockwell Hardness of Plastics and Electrical Insulating Materials

ASTM D789 - Standard Test Methods for Determination of Solution Viscosities of Polyamide (PA)

ASTM D1822 - Standard Test Method for Tensile-Impact Energy to Break Plastics and Electrical Insulating Materials

ASTM D3418 - Standard Test Method for Transition Temperatures and Enthalpies of Fusion and Crystallization of Polymers by Differential Scanning Calorimetry

ASTM D5989 - Standard Specification for Extruded and Monomer Cast Shapes Made from Nylon (PA)

(Copies of these documents are available online at www.astm.org.)

2.4 Order of precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

## 3. REQUIREMENTS

3.1 First article. When specified (see 6.2), a sample shall be subjected to first article inspection in accordance with 4.2.

3.2 Material.

3.2.1 Composition. The material shall consist of unmodified virgin plastic polyamide, nylon 66, poly (hexamethylenediamine), with 1±¼ percent by weight of molybdenum disulfide (technical fine grade), and shall be gray-black in color.

3.2.2 Environmentally preferable materials. Environmentally preferable materials should be used to the maximum extent possible, provided that the material meets or exceeds the operational and maintenance requirements, and promotes economically advantageous life cycle costs.

3.2.3 Prohibited materials. Recycled plastic materials shall not be used.

3.2.4 Material safety data sheet. DELETED

3.2.5 Toxicity. When evaluated in accordance with 4.5 (the Health Hazard Assessment [HHA]), the item shall have no adverse effect on the health of personnel when used for its intended purpose (see 4.5 and 6.5).

3.3 Dimensions and tolerances. Dimensions and tolerances shall be as specified (see 6.2). Material shall be provided as either an extruded rod stock shape or extruded plate stock shape.

3.4 Melting point. The melting point of the material shall be in the range 482 to 509 °F (250 to 265 °C).

3.5 Relative viscosity. The relative viscosity shall be a minimum of 190.

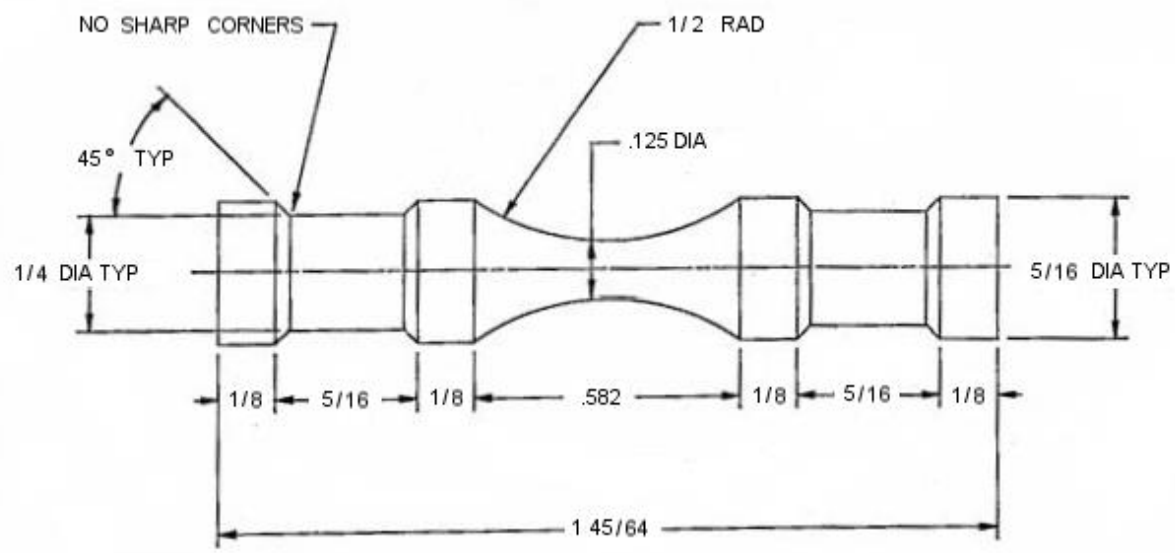
3.6 Percent molybdenum disulfide. The percentage of molybdenum disulfide in the material shall be 1±¼ percent.

3.7 Dimensional stability. The percent change in dimensions shall be a maximum of 0.2 percent for the material.



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3.8 Tensile impact. When tested in accordance with 4.9, the minimum average tensile impact energy for round tensile-impact specimens depicted on figure 1 shall be 80 foot-pounds per square inch, with no individual specimen less than 70 foot-pounds per square inch. Alternatively, Type S rectangular specimens in accordance with ASTM D1822 may be used: the minimum average tensile impact energy value shall be 60 foot-pounds per square inch and individual specimen values shall not be less than 50 foot-pounds per square inch. If unacceptable individual test results show evidence of poor specimen preparation or improper test method, the test may be repeated using a double sample, and the test report will include a statement explaining the reason for the retest.




NOTES:

1. All dimensions are shown in inches.
2. Allowable tolerances shall be ±0.005 for decimals and ±1/64 for fractions.
3. Surface roughness shall not exceed 125 microinches in accordance with ASME B46.1.

FIGURE 1. Round cylindrical tensile impact test specimen.

3.9 Hardness. The Rockwell "M" Hardness shall be a minimum of 75.

3.10 Workmanship. Stock shapes supplied to this specification shall have a smooth finish and shall be free of voids, pin holes, inclusions of foreign matter, or other defects.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.2).

b. Conformance inspection (see 4.3).

4.2 First article inspection. First article inspection shall be performed on material used for ball valve seat seals and thrust washers (nylon 66 with molybdenum disulfide) in petroleum-based hydraulic fluid systems when a first article sample is required (see 3.1). This inspection shall include the sampling requirements of 4.4 and the tests of 4.5 through 4.11 (see table I).



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TABLE I. Tests.

| Item | First article testing required | Conformance testing required | Requirement paragraph | Test method paragraph | No. of test samples per extruded lot | No. of test specimens per extruded lot |
|---|---|---|---|---|---|---|
| Toxicity | X | | 3.2.5 | 4.5 | NA | NA |
| Melting Point | X | X | 3.4 | 4.6 | 1 | 2 |
| Relative Viscosity | X | X | 3.5 | 4.7 | 1 | 2 |
| % Molybdenum Disulfide | X | X | 3.6 | 4.8 | 2 | 4 |
| Tensile Impact | X | X | 3.8 | 4.9 | 2 | 10 |
| Rockwell Hardness | X | X | 3.9 | 4.10 | 2 | 4 |
| Dimensional Change | X | X | 3.7 | 4.11 | 2 | 4 |

4.3 Conformance inspection. Conformance inspection shall include the sampling requirements of 4.4 and the tests of 4.6 through 4.11 (see table I).

4.4 Sampling.

4.4.1 Lot. For purposes of sampling, a lot (extruded rod or plate) shall consist of the final heat treat of the product, annealed without interruption at one time and consisting of product from a single resin and concentrate lot and a single size and shape extrusion run.

4.4.2 Sampling requirements. The contractor shall select a production sample sufficiently large enough to permit the performance of all tests. The number of samples shall be in accordance with table I. Sample locations shall be selected so that they will be representative of the total lot. Duplicate specimens shall be made from each sample. Where referenced ASTM specifications and this specification do not agree on the number of test specimens, this specification shall be used.

4.4.3 Noncompliance. If a sample fails to pass its inspections, the lot shall be rejected as specified (see 6.2 and 6.6).

4.5 Toxicity. An HHA will be conducted to ensure conformance to 3.2.5, as specified (see 6.2). The Navy and Marine Corps Public Health Center (NMCPHC) will evaluate the material using data provided by the manufacturer/distributor to the NMCPHC (see 3.2.5 and 6.5).

4.6 Melting point. The melting point for the material shall be determined in accordance with ASTM D3418.

4.7 Relative viscosity. The relative viscosity shall be determined in accordance with ASTM D789. Prior to the test, specimens shall be dried in accordance with the testing requirements for tensile strength and elongation in L-P-410. In addition to the viscometers referenced in ASTM D789, the Cannon-Fenske viscometer may also be used.

4.8 Percent molybdenum disulfide. Percent molybdenum disulfide shall be determined in accordance with Appendix A or other NAVSEA approved method.

4.9 Tensile impact. Tensile impact energy shall be determined in accordance with ASTM D1822 (using round test specimens per figure 1 or Type S rectangular samples per ASTM D1822). Prior to the test, specimens shall be dried in accordance with the testing requirements for tensile strength and elongation in L-P-410.



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4.10 Hardness. Rockwell "M" hardness shall be determined per ASTM D785. Determinations shall be performed on the dimensional change test specimens of 4.11 prior to performing the dimensional test.

4.11 Dimensional change. The percent change in dimensions following heating shall be determined in accordance with dimensional stability of ASTM D5989.

4.12 Certificate of compliance. DELETED

4.13 Test reports. Test results shall be reported as specified (see 6.2) and shall cite the relevant classification lot number and the resin and concentrate lot number to assure traceability.

5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. This specification covers nylon 66 with molybdenum disulfide material for use as ball valve seats and thrust washers in petroleum-based hydraulic fluid systems. This specification is military-unique due to a lack of commercial specifications that have these requirements and its use on submarines and surface ships. This material is not recommended or intended for use in water. Use of this material in other fluids (such as synthetic hydraulic fluids or compressed gas) should be evaluated and approved by the system technical authority.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Amount of material required.
c. When first article is required (see 3.1 and 6.4).
d. Dimensions and tolerances (see 3.3).
e. Instructions for noncompliant samples (see 4.4.3).
f. Requirements for an HHA (see 3.2.5 and 4.5).
g. Test reports (see 4.13).
h. Packaging requirements (see 5.1).
i. Requirements for certificate of compliance (see 6.3).
j. Material safety data sheet requirements (see 6.7.1).

6.3 Certificate of compliance. A certificate of compliance verifying conformance to this specification should be prepared (see 6.2). The certificate should reference the order number and contract number. The certificate should cite the relevant classification lot number and the resin and concentrate lot number to assure traceability. Test results for each of the six conformance tests in table I should be referenced on the certificate.

6.4 Formulation changes. Any change in basic ingredients or manufacturing processes which would affect compliance with this specification should be reported to both the contracting activity and NAVSEA. Any formulation change not disclosed to NAVSEA will result in immediate product rejection.



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6.5 Toxicity evaluation. The NMCPHC requires sufficient information to permit an HHA of the product. Upon completion of the HHA, a copy will be provided by the NMCPHC to the Government for evaluation. The HHA process is described on the NMCPHC's website, http://www.med.navy.mil/sites/nmcphc/industrial-hygiene/Pages/health-hazard-assessment.aspx.

6.6 Noncompliant samples. The manufacturer should notify the cognizant inspection activity of sample failure to pass conformance inspection and should take corrective action on the materials or processes, or both, as warranted. Acceptance and shipment of the product should be discontinued until corrective action, acceptable to the inspection activity, has been taken. After the corrective action has been taken, conformance inspections should be repeated on the new lot. In the event of failure after re-inspection, information concerning the failure should be furnished to the cognizant inspection activity.

6.7 Subject term (key word) listing.

Hydraulic system

Polyamide

6.7.1 Material safety data sheet. The contracting activity should be provided a Material Safety Data Sheet (MSDS) prepared in accordance with FED-STD-313 at the time of the contract award. In order to obtain the MSDS, FAR clause 52.223-3 must be in the contract. The MSDS should be included with the initial shipment of the material covered by this specification; the MSDS number should be identified on all shipments, and the MSDS should be available electronically.

6.8 Amendment notations. The margins of this specification are marked with vertical lines to indicate modifications generated by this amendment. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations.



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# PROCEDURE TO DETERMINE PERCENT MOLYBDENUM DISULFIDE

## A.1 SCOPE

A.1.1 Scope. This test, which can be used for the determination of molybdenum disulfide ($MoS_2$), is a modification of the testing requirements for molybdenum disulfide in L-P-410 (which references the testing requirements for carbon content of the same specification). The method is based on the hydrolysis of nylon with hydrochloric acid and the subsequent separation of the $MoS_2$ residue by filtration. This appendix is not a mandatory part of the specification. The information contained herein is intended for guidance only.

## A.2 TEST APPARATUS

A.2.1 Equipment. Test equipment includes the following:

a. Oven capable of maintaining 100±5 °C (212±9 °F).
b. Analytical balance accurate to 0.1 milligram.
c. A filter holder.
d. 0.5 micrometer TFE filters for the filter holder in A.2.1.c.
e. 1000 milliliter vacuum flask.
f. Source of vacuum (if a pump is used, a trap must be placed in a series with the system in order to protect the pump).
g. Graduated cylinders: 500 milliliter, 1000 milliliter, and 2000 milliliter.
h. 1 gallon capacity polyethylene containers (for mixing reagents).
i. Laboratory grinder (capable of grinding the sample to ⅛-inch diameter particles or smaller).
j. Supply of 500 milliliter flat bottom flasks, with 24/10 ground glass joints.
k. 4 bulb or reflux condensers, with 24/10 ground glass joints.
l. Hot plate.
m. Supply of aluminum weighing dishes.
n. Small pair of forceps.
o. Laboratory fume hood.
p. Supply of distilled or demineralized water.
q. Desiccators with desiccant.

A.2.2 Reagents.

a. Acetone (commercial grade).
b. Formic acid, 88 to 90 percent (technical grade).
c. Hydrochloric acid solution, 1:1 (approx. 6N) – (reagent grade).
d. Hydrochloric acid solution, 1:3 (approx. 3N) – (reagent grade).



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A.3 PROCEDURE

A.3.1 Steps.

a. Grind enough material to give at least 10 grams of the ground product (nylon resin can be tested "as received").
b. Weigh 5.0±0.001 grams of sample to the nearest 0.1 milligram and record the weight. Place the sample into a clean 500-milliliter flask.
c. Under a fume hood, add 200 milliliters of 6N hydrochloric acid (HCl) to the sample. Attach the flask to a condenser and place on the hot plate. Circulate cooling water through the condenser.
d. The mixture is heated just to boiling. Continue boiling until digestion is complete. This usually occurs within a period of one hour.
e. Weigh a TFE filter to the nearest 0.1 milligram, and record the weight. Place the filter in the filter holder as per the instructions supplied with the holder. The assembly is then placed on top of the vacuum flask.
f. After the digestion (hydrolysis) period, the solution is diluted with 200 milliliters of boiling, distilled or demineralized water. If hydrolysis is complete, no precipitation should be visible at this point. This solution is again taken just to boiling and held there for 15 minutes.
g. Remove the flask from the hot plate and allow the $MoS_2$ to settle for 15 to 20 minutes. This settling step allows the bulk of the separation to be accomplished by decantation into the filter assembly and greatly reduces filtration time.
h. Start pulling a vacuum on the filter assembly.
i. While spraying the filter with acetone, pour the warm solution slowly into the filter assembly. As the liquid is drawn into the flask and the level in the filter holder drops, more solution can be added. Continue this procedure until all of the solution has been poured from the flask.
j. Rinse the flask with approximately 150 milliliters of hot 3N HCl to remove any remaining residue. While spraying the filter with acetone, pour the rinsing solution onto the filter.
k. The residue left on the filter is then rinsed with 100 milliliters of hot (not boiling) formic acid. This is done while spraying with acetone.
l. A final acetone rinse is then made.
m. Carefully remove the filter from the filter holder with forceps and place in an aluminum dish. Place the dish in an oven which has been pre-heated to 100±5 °C (212±9 °F) and dry for 1 hour.
n. Remove the dish with the filter and residue from the oven and immediately place in a desiccator to cool.
o. Weigh the filter and residue to the nearest 0.1 milligram and record.

A.4 CONTROL SAMPLE

A.4.1 Use of control sample. Since this test will detect any filler not hydrolyzable in the nylon, it will be necessary to test the base nylon which was used, before the $MoS_2$ was added, whenever possible. Should this control sample contain any material which does not hydrolyze, this percent residue value must be subtracted from that found in the test samples (see A.5). Follow the same testing procedure of A.3.



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A.5 CALCULATIONS

A.5.1 Non-hydrolyzable solids. Calculate the percent non-hydrolyzable solids as follows:

Percent non-hydrolyzable solids = ((A-B)/C x 100) – D

A = Weight of filter and residue after final drying

B = Weight of clean filter

C = Weight of sample

D = Percent of non-hydrolyzable solids of control sample (see A.4)

A.5.2 Reporting. Report the individual and average values obtained as percent non-hydrolyzable solids ($MoS_2$ percentage) to the nearest 0.01 percent. If a "control" is not available, the result should be reported as "uncorrected"; for example: 1.05 percent (uncorrected).

A.5.3 Reproducibility. Testing should be done in duplicate, with the results reproducing to within 5 percent of their average.



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